

Cancer Biology

Integrin $\beta 4$ promotes DNA damage drug resistance in triple-negative breast cancer via TNFAIP2/IQGAP1/Rac1

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Huan Fang, Wenlong Ren, Qiuxia Cui, Huichun Liang, Chuanyu Yang, Wenjing Liu, Xinye Wang, Xue Liu, Yujie Shi , Jing Feng , Ceshi Chen 

Key Laboratory of Animal Models and Human Disease Mechanisms, Kunming Institute of Zoology, Chinese Academy of Sciences, Kunming, Yunnan, China. • Kunming College of Life Sciences, University of Chinese Academy Sciences, Kunming, Yunnan, China. • School of Life Science, University of Science & Technology of China, Hefei, Anhui, China. • Affiliated Hospital of Guangdong Medical University, Zhanjiang, Guangdong, China. • Department of Breast Surgical Oncology, National Cancer Center/National Clinical Research Center for Cancer/Cancer Hospital & Shenzhen Hospital, Chinese Academy of Medical Sciences and Peking Union Medical College, Shenzhen, Guangdong, China. • Shanghai University of Medicine & Health Sciences, Affiliated Sixth People's Hospital South Campus, Shanghai, China. • Department of Pathology, Henan Provincial People's Hospital, Zhengzhou University, Zhengzhou, Henan, 450003, China. • The Second Affiliated Hospital of the Chinese University of Hong Kong (Shenzhen), Shenzhen, 518172, China. • Academy of Biomedical Engineering, Kunming Medical University, Kunming 650500, China. • The Third Affiliated Hospital, Kunming Medical University, Kunming 650118, China.

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Abstract

Summary

Anti-tumor drug resistance is a challenge for triple-negative breast cancer treatment. Our previous work demonstrated that TNFAIP2 activates Rac1 to promote triple-negative breast cancer cell proliferation and migration. However, the mechanism by which TNFAIP2 activates Rac1 is unknown. In this study, we found that TNFAIP2 interacts with IQGAP1 and Integrin $\beta 4$. Integrin $\beta 4$ activates Rac1 through TNFAIP2 and IQGAP1 and confers DNA damage drug resistance in triple-negative breast cancer. These results indicate that the Integrin $\beta 4$ /TNFAIP2/IQGAP1/Rac1 axis provides potential therapeutic targets to overcome DNA damage drug resistance in triple-negative breast cancer.

eLife assessment

This study presents a **valuable** finding that TNFAIP2 interacts with IQGAP1 and Integrin β 4. Moreover, Integrin β 4 activates Rac1 through TNFAIP2 and IQGAP1 and confers drug resistance in triple-negative breast cancer. The evidence supporting the claims of the authors is **solid**. The work will be of interest to medical biologists working on breast cancer or drug resistance.

Introduction

Breast cancer is the most common malignant tumor in women, and the morbidity and mortality rates of female malignant tumors are ranked first in the world⁽¹⁾. Although the diagnosis and treatment of breast cancer has entered the era of molecular typing, 35% of breast cancers still experience recurrence, metastasis and treatment failure⁽²⁾. According to the expression of estrogen receptor (ER α), progesterone receptor (PR) and human epidermal growth factor receptor (HER2), breast cancer is divided into ER/PR-positive, HER2-positive and triple-negative breast cancer (TNBC)⁽³⁾. For ER/PR-positive and HER2-positive breast cancer, endocrine therapies such as tamoxifen and anti-HER2 targeted therapy such as trastuzumab have achieved good efficacy. Targeted drugs for TNBC patients with BRCA1/2 mutations include two PARP inhibitors, olaparib and talazoparib. These targeted drugs cannot fully meet the clinical needs of patients with various TNBC subtypes⁽⁴⁾. Currently, DNA damage chemotherapy drugs, including epirubicin and cisplatin, are widely used for TNBC treatment.

TNBC often recurs and metastasizes due to the development of chemoresistance, although it is initially responsive to chemotherapeutic drugs⁽⁵⁾. Chemoresistance severely impacts the clinical outcomes of patients. Tumor cells become resistant to chemotherapeutic agents through several mechanisms, such as improving DNA damage repair, changing the intracellular accumulation of drugs, or increasing anti-apoptotic mechanisms⁽⁶⁾. Therefore, characterization of the underlying molecular mechanisms by which resistance occurs will provide opportunities to develop precise therapies to enhance the efficacy of standard chemotherapy regimens^{(7), (8)}.

TNFAIP2 is abnormally highly expressed in a variety of tumor cells, including TNBC⁽⁹⁾, nasopharyngeal carcinoma⁽¹⁰⁾, malignant glioma⁽¹¹⁾, uroepithelial carcinoma⁽¹²⁾ and esophageal squamous cell carcinoma⁽¹³⁾, and is associated with poor prognosis. Our previous work^{(9), (14)} showed that TNFAIP2, as a KLF5 downstream target protein, can interact with Rac1⁽¹⁵⁾, a member of the Rho small GTP enzyme family, and activate Rac1 to alter the cytoskeleton, thereby inducing filopodia and lamellipodia formation and promoting the adhesion, migration and invasion of TNBC cells. After activation, Rac1 can activate AKT, PAKs, NADPH oxidase and other related signaling pathways to promote cell survival, proliferation, adhesion, migration and invasion⁽¹⁶⁾. Activation of Rac1 can reduce the therapeutic response to trastuzumab in breast cancer and increase the resistance of TNBC cells to paclitaxel⁽¹⁷⁾, but the specific mechanism of action is not completely clear.

Rac1 has been shown to play an important role in DNA damage repair. Activated Rac1 can promote the phosphorylation of the DNA damage response-related molecules ATM/ATR, CHK1/2 and H2AX by activating the activity of protein kinases such as ERK1/2, JNK and p38^{(18), (19)}, thus improving the DNA damage repair ability and inhibiting tumor cell

apoptosis^{(20)–(22)}. Rac1 also promotes aldolase release and activation by changing the cytoskeleton and activates the ERK pathway to increase the pentose phosphate pathway to promote nucleic acid synthesis, providing more raw materials for DNA damage repair^{(23), (24)}. At the same time, the interaction of Rac1 and PI3K promoted AKT phosphorylation and glucose uptake^{(25), (26)}. Therefore, Rac1 is well established to promote the chemoresistance of breast cancer by promoting DNA damage repair.

Integrin $\beta 4$ (ITGB4) is a major component of hemidesmosomes and a receptor molecule of laminin. Studies have shown that laminin-5 interacts with ITGB4 to activate Rac1 activity and promote cell migration⁽²⁷⁾ and polarization⁽²⁸⁾ by altering the cytoskeleton. Since ITGB4-positive cancer stem cell (CSC)-enriched mesenchymal cells were found to reside in an intermediate epithelial/mesenchymal phenotypic state, ITGB4 can be used to enable stratification of mesenchymal-like TNBC cells⁽²⁹⁾. In addition, the expression of ITGB4 on ALDH^{high} breast cancer and head and neck cancer cells was significantly greater than that on ALDH^{low} cells, proving the effects that ITGB4 targets on both bulk and CSC populations⁽³⁰⁾. Furthermore, ITGB4-overexpressing TNBC cells provided cancer-associated fibroblasts (CAFs) with ITGB4 proteins via exosomes, and ITGB4-overexpressing CAF-conditioned medium promoted the proliferation, epithelial-to-mesenchymal transition, and invasion of breast cancer cells⁽³¹⁾. ITGB4 also promotes breast cancer cell resistance to tamoxifen-induced apoptosis by activating the PI3K/AKT signaling pathway and promotes breast cancer cell resistance to anoikis by activating Rac1⁽³²⁾. However, how ITGB4 activates Rac1 is not completely clear.

Rac1 activity is regulated by guanylate exchange factors (GEFs), GTPase activation proteins (GAPs), and guanine separation inhibitors (GDIs)⁽³³⁾. GAPs typically provide the necessary catalytic groups for GTP hydrolysis, but not all GAPs function as hydrolases. IQGAP1 lacks an arginine in the GTPase binding domain and cannot exert the hydrolysis effect of GAPs⁽³⁴⁾. IQGAP1 can increase the activity of Rac1 and CDC42^{(35), (36)}.

In this study, we demonstrated that TNFAIP2 interacts with IQGAP1 and ITGB4. ITGB4 promotes TNBC drug resistance via the TNFAIP2/IQGAP1/Rac1 axis by promoting DNA damage repair. Our results suggest that ITGB4 and TNFAIP2 might serve as promising therapeutic targets for TNBC.

Results

TNFAIP2 promotes TNBC DNA damage drug resistance

To explore the functional significance of TNFAIP2 in TNBC drug resistance, we constructed stable TNFAIP2 overexpression and TNFAIP2 knockdown HCC1806 and HCC1937 cells. As shown in [Figure 1A–E](#), overexpression of TNFAIP2 significantly increased cell viability when treated with EPI and BMN. Additionally, knockdown of TNFAIP2 significantly decreased cell viability when treated with EPI and BMN ([Fig. 1F–J](#)). We then examined the effects of TNFAIP2 on DNA damage repair and found that TNFAIP2 promotes DNA damage repair in response to EPI and BMN. TNFAIP2 overexpression decreased the protein expression levels of γ H2AX, a marker of DNA damage, and cleaved-PARP, a marker of apoptosis ([Fig. 1K](#)). Additionally, knockdown of TNFAIP2 significantly increased γ H2AX and cleaved-PARP protein expression levels in response to EPI and BMN in both cell lines ([Fig. 1L](#)).

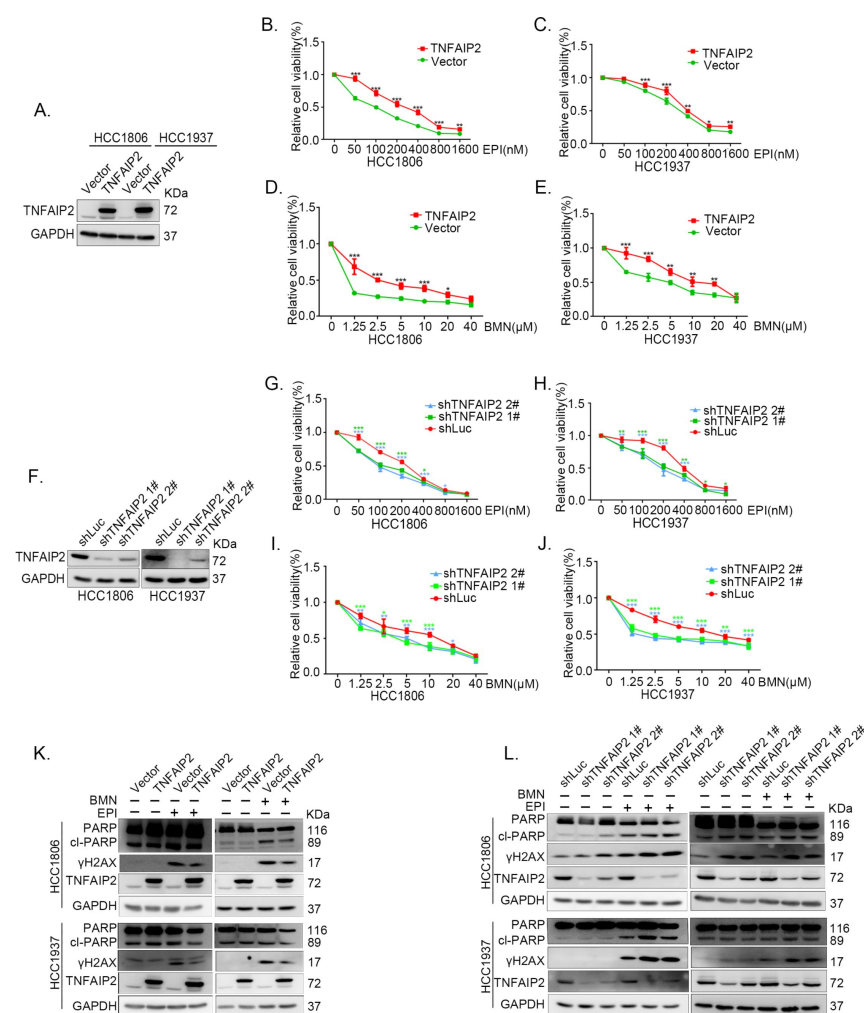


Figure 1.

TNFAIP2 promotes TNBC DNA damage drug resistance

(A-E) Stable TNFAIP2 overexpression in HCC1806 and HCC1937 cells significantly increased cell viability in the presence of EPI (0-1.6 μM) or BMN (0-40 μM) treatment for 48h, as measured by the SRB assay. TNFAIP2 protein expression was detected by WB.

(F-J) Stable TNFAIP2 knockdown in HCC1806 and HCC1937 cells significantly decreased cell viability in the presence of EPI (0-1.6 μM) or BMN (0-40 μM) treatment for 48h, as measured by the SRB assay. TNFAIP2 protein expression was detected by WB.

(K) TNFAIP2 promoted DNA damage repair in the presence of EPI and BMN. HCC1806 and HCC1937 cells stably overexpressing TNFAIP2 were treated with 400 or 800 nM EPI for 48h and 10 μM BMN for 24h, respectively. TNFAIP2, γH2AX, and PARP protein expression was detected by WB.

(L) TNFAIP2 knockdown increased DNA

damage in the presence of EPI and BMN. Stable TNFAIP2 knockdown cells were treated with 400 or 800 nM EPI for 24 or 48h and 2.5 μM BMN for 24h. TNFAIP2, γH2AX, and PARP protein expression was detected by WB.

The function of TNFAIP2 was further validated by using two other DNA damage drugs, DDP and AZD (Fig. S1A-L). These results suggested that TNFAIP2 enhances TNBC cell drug resistance by promoting DNA damage repair.

TNFAIP2 confers TNBC drug resistance *in vivo*

To test whether TNFAIP2 also decreases the sensitivity of TNBC cells to EPI and BMN *in vivo*, we performed animal experiments in nude mice. HCC1806 cells with stable TNFAIP2 knockdown were orthotopically inoculated into the fat pad of 7-week-old female mice (n=8 or 12/group). When the tumor mass reached approximately 50 mm³, each group was divided into two subgroups to receive either EPI (2.5 mg/kg, twice a week) or vehicle control for 23 days and either BMN (1 mg/kg, twice a week) or vehicle control for 29 days. We observed that depletion of TNFAIP2 suppressed breast cancer cell growth *in vivo*. This is consistent with our previous report⁽⁹⁾. More importantly, TNFAIP2 depletion further decreased tumor volume when mice were treated with EPI and BMN (Fig. 2A-F). Meanwhile, BMN treatment had no effect on the body weight of mice (Fig. S2F). Consistently, EPI and DDP generated similar results but decreased mouse body weight due to their high toxicity

(Fig.S2D-E). These results suggest that inhibition of TNFAIP2 expression can overcome HCC1806 breast cancer cell drug resistance in animals.

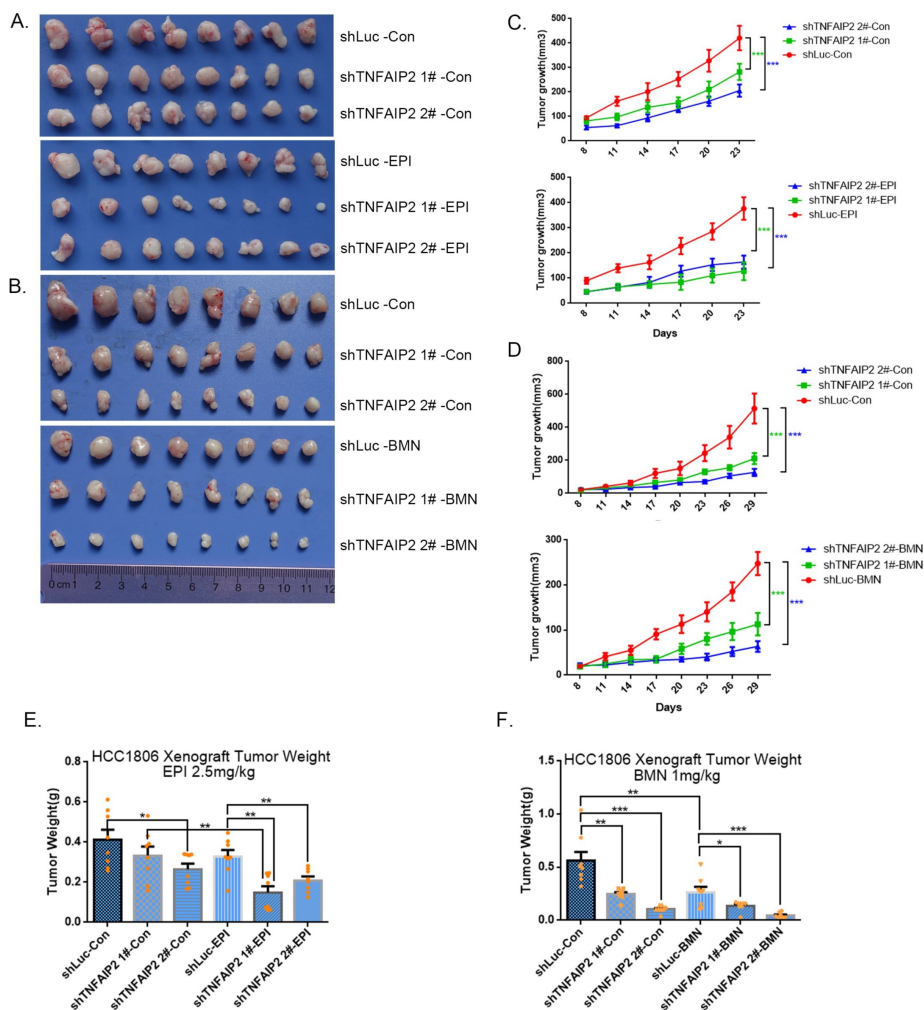


Figure2.

TNFAIP2 confers TNBC drug resistance *in vivo*

(A-F) TNFAIP2 knockdown increased the sensitivity of HCC1806 breast cancer cells to EPI and BMN *in vivo*. HCC1806 cells with stable TNFAIP2 knockdown were transplanted into the fat pad of 7-week-old female nude mice. When the average tumor size reached approximately 50 mm³ after inoculation, mice in each group were randomly divided into two subgroups (n=4/group) to receive EPI (2.5 mg/kg), BMN (1 mg/kg) or vehicle control for 23 or 29 days (A-B). Tumor size was measured twice a week (C-D), and tumor masses were collected and weighed at the end of the experiments (E-F). *: $P < 0.05$, **: $P < 0.01$, t -test.

TNFAIP2 promotes TNBC drug resistance and DNA damage repair via Rac1

Since chemotherapeutic agents and PRAP inhibitors induce DNA damage directly or indirectly, DNA damage repair ability profoundly affects the sensitivity of cancer cells to these therapies^{(37), (38)}. Since TNFAIP2 can activate Rac1, a well-known drug resistance protein, we investigated whether TNFAIP2 induces chemotherapeutic resistance through Rac1. We found that Rac1 knockdown rescued the effects of TNFAIP2 overexpression-induced drug resistance to EPI and BMN in HCC1806 and HCC1937 cells (Fig. 3A-F). We also found that γ H2AX and cleaved-PARP protein levels were up-regulated again in Rac1 knockdown and TNFAIP2-overexpressing HCC1806 and HCC1937 cells in response to EPI and BMN (Fig. 3G-J). We obtained similar results by using DDP and AZD treatment (Fig. 3SA-J). Collectively, these results suggest that TNFAIP2 promotes DNA damage repair and drug resistance via Rac1.

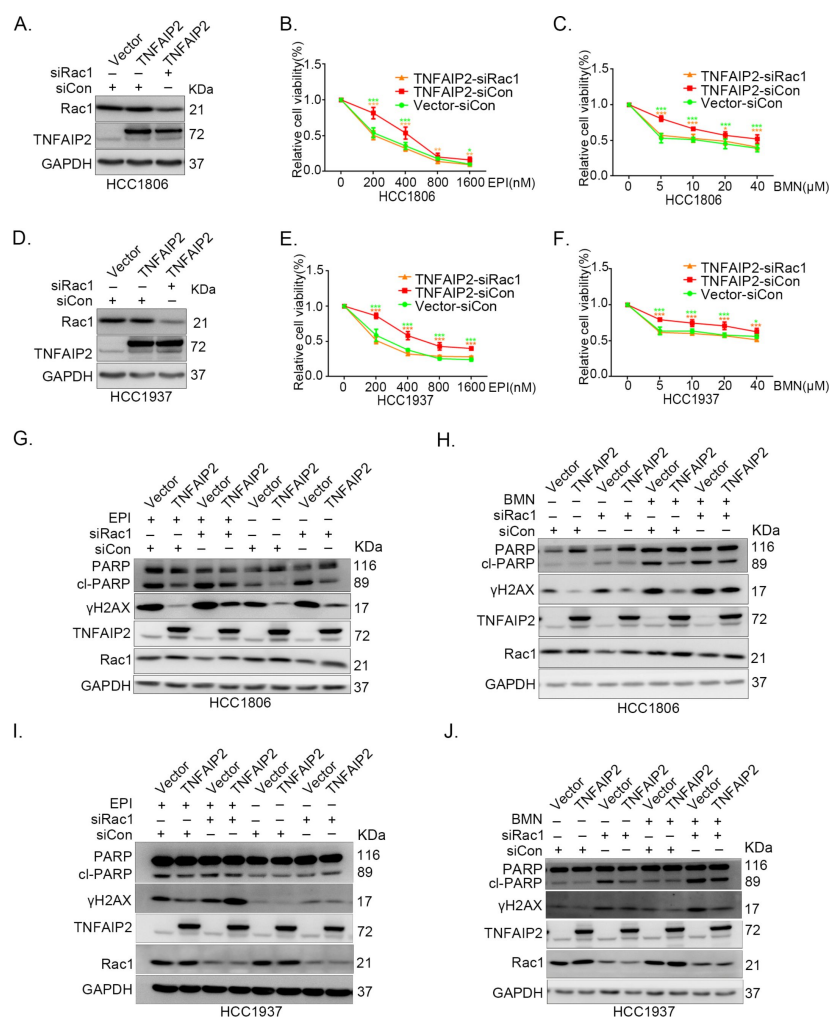


Figure 3.

TNFAIP2 promotes TNBC drug resistance and DNA damage repair via Rac1

(A-F) Rac1 knockdown abolished TNFAIP2-induced TNBC resistance to EPI and BMN. HCC1806 (A-C) and HCC1937 (D-F) cells with stable TNFAIP2 overexpression were transfected with Rac1 or control siRNA, followed by treatment with EPI (0-1600 nM) and BMN (0-40 μ M) for 48 or 72 h, respectively. Cell viability was measured by the SRB assay. Protein expression levels were analyzed by WB.

(G-J) Rac1 depletion abolished TNFAIP2-induced DNA damage decrease in response to EPI and BMN. HCC1806 (G-H) and HCC1937 (I-J) cells with stable TNFAIP2 overexpression were transfected with Rac1 or control siRNA, followed by treatment with EPI (400 or 800 nM) and BMN (10 μ M) for 24h, respectively. Protein expression levels were analyzed by WB.

IQGAP1 mediates Rac1 activation by TNFAIP2 and promotes TNBC drug resistance

To characterize the mechanism by which TNFAIP2 activates Rac1, we performed an IP-MS experiment. We found that TNFAIP2 interacts with IQGAP1 (Fig. 4A). To validate whether TNFAIP2 interacts with IQGAP1, we constructed HCC1806 cells with stable Flag-TNFAIP2 overexpression and collected Flag-tagged TNFAIP2 cell lysates for immunoprecipitation assays using Flag-M2 beads (Fig. S4A). We performed immunoprecipitation using anti-IQGAP1 antibody and found that endogenous IQGAP1 protein interacted with endogenous TNFAIP2 protein in HCC1806 cells (Fig. 4B). Next, we mapped the regions of TNFAIP2 and IQGAP1 proteins responsible for the interaction by generating a series of Flag-TNFAIP2 deletion mutants and transfected them into HEK293T cells together with full-length IQGAP1. Then, we performed immunoprecipitation assays using Flag-M2 beads (Figure S4B), and the N-terminus (1-79 aa) of the TNFAIP2 protein interacted with IQGAP1. To explore the function of IQGAP1 in TNBC drug resistance, we knocked down IQGAP1 in HCC1806 and HCC1937 cells. As shown in Figure 4C-G, knockdown of IQGAP1 significantly decreased cell viability in the presence of EPI and BMN in both cell lines. We also examined the effects of IQGAP1 on DNA damage repair and found that IQGAP1 promotes DNA damage repair. IQGAP1 knockdown increased γ H2AX and cleaved-PARP protein expression levels when HCC1806 and HCC1937 cells were treated with EPI and BMN (Fig. 4H). Next, we found that IQGAP1 knockdown abrogated the effects of TNFAIP2 overexpression on resistance to EPI and BMN

(Fig. 4I-K, Fig.S4C-E). We also found that γ H2AX and cleaved-PARP protein levels were up-regulated in IQGAP1 knockdown and TNFAIP2-overexpressing HCC1806 and HCC1937 cells (Fig. 4L). In addition, we found that the TNFAIP2 overexpression-induced increase in Rac1 activity was abolished by IQGAP1 knockdown (Fig. 4M).

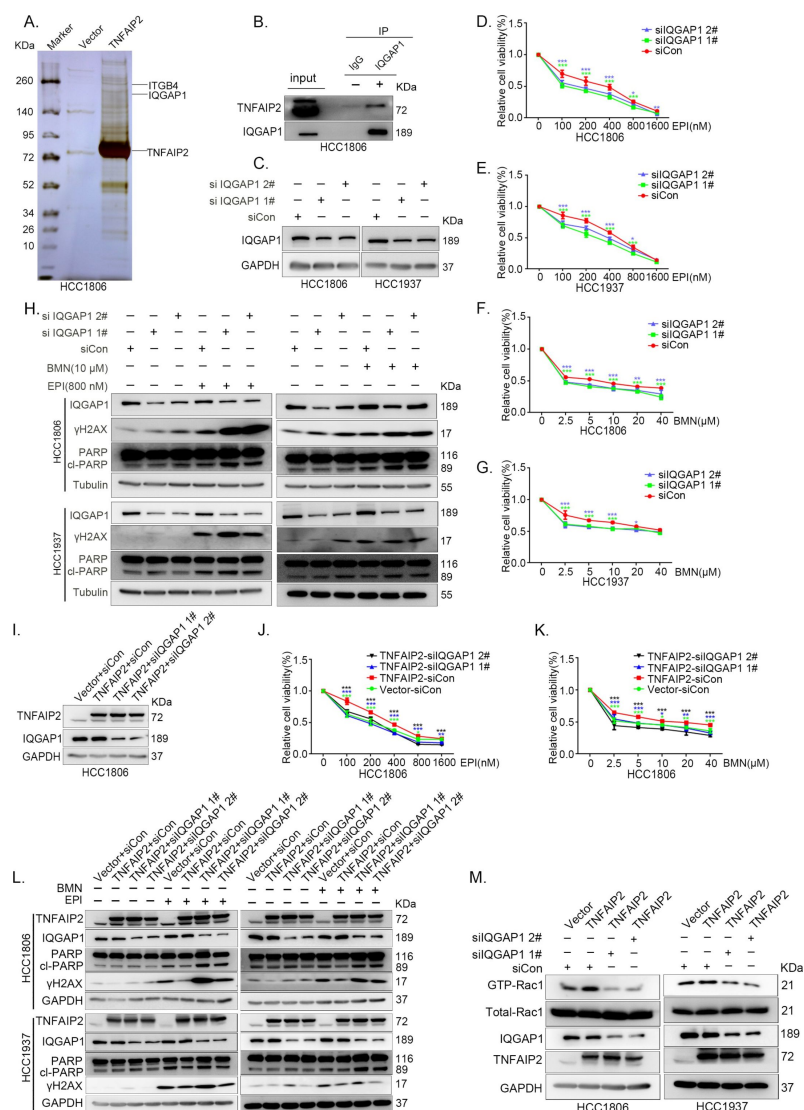


Figure 4.

IQGAP1 mediates Rac1 activation by TNFAIP2 and promotes TNBC drug resistance

(A) The IP-MS result of TNFAIP2 in HCC1806 cells.

(B) Endogenous TNFAIP2 interacts with IQGAP1 in HCC1806 cells. Endogenous TNFAIP2 protein was immunoprecipitated using anti-IQGAP1 antibody. Immunoglobulin (Ig)G served as the negative control. Endogenous TNFAIP2 was detected by WB.

(C-G) IQGAP1 knockdown in HCC1806 and HCC1937 cells significantly decreased cell viability in the presence of EPI (0-1600 nM) and BMN (0-40 μ M), as measured by the SRB assay. IQGAP1 protein expression was detected by WB.

(H) IQGAP1 knockdown in HCC1806 and HCC1937 cells increased DNA damage of EPI and BMN. HCC1806 and HCC1937 cells with IQGAP1 knockdown were treated with 800 nM EPI for 24h and 10 μ M BMN for 24h, respectively. ITGB4, γ H2AX, and PARP protein expression was detected by WB.

(I-K) IQGAP1 knockdown abolished TNFAIP2-conferred resistance to EPI and BMN. HCC1806 cells with stable TNFAIP2 overexpression were

transfected with IQGAP1 or control siRNA, followed by treatment with EPI (0-1600 nM) and BMN (0-40 μ M) for 48 or 72 h, respectively. Cell viability was measured by the SRB assay. IQGAP1 protein expression was detected by WB.

(L) IQGAP1 knockdown abolished TNFAIP2-conferred resistance to EPI and BMN. HCC1806 and HCC1937 cells with stable TNFAIP2 overexpression were transfected with IQGAP1 or control siRNA, followed by treatment with EPI (800 nM) and BMN (10 μ M) for 24h, respectively. Protein expression levels were analyzed by WB.

(M) IQGAP1 knockdown abolished TNFAIP2-conferred Rac1 activation. HCC1806 and HCC1937 cells with stable TNFAIP2 overexpression were transfected with IQGAP1 or control siRNA. GTP-Rac1 levels were assessed using PAK-PBD beads.

ITGB4 interacts with TNFAIP2 and promotes TNBC drug resistance and DNA damage repair

In addition to IQGAP1, TNFAIP2 may interact with ITGB4 (Fig. 4A). To validate whether TNFAIP2 interacts with ITGB4, we immunoprecipitated exogenous Flag-tagged TNFAIP2 proteins from HCC1806 cells by using Flag-M2 beads and detected endogenous ITGB4 proteins (Fig. 5A). To further confirm the protein-protein interaction between endogenous TNFAIP2 and ITGB4 proteins, we collected HCC1806 cell lysates and performed immunoprecipitation using an anti-TNFAIP2/ITGB4 antibody and found that endogenous TNFAIP2/ITGB4 protein interacted with endogenous ITGB4/TNFAIP2 protein (Fig. 5B-C). We further mapped the regions of TNFAIP2 and ITGB4 proteins responsible for the interaction (Fig. S5I) by generating a series of Flag-TNFAIP2/GST-fused TNFAIP2 deletion mutants and transfected them into HEK293T cells together with full-length GST-fused ITGB4/ITGB4. Then, we performed immunoprecipitation assays using Flag-M2 beads and glutathione beads. As shown in Figure S5J-K, the N-terminus (218-287 aa) of the TNFAIP2 (TNFAIP2-S-N1-3) protein interacted with ITGB4. To map the domains of ITGB4 that interact with TNFAIP2, we transfected Flag-tagged full-length TNFAIP2 into HEK293T cells with full-length or truncated ITGB4. We found that the C-terminus (710-740 aa) of the ITGB4 protein interacted with TNFAIP2 (Fig. S5L-M). Taken together, these results suggest that TNFAIP2 interacts with ITGB4 and that their interaction is mediated through the N-terminus of TNFAIP2 and the C-terminus of ITGB4.

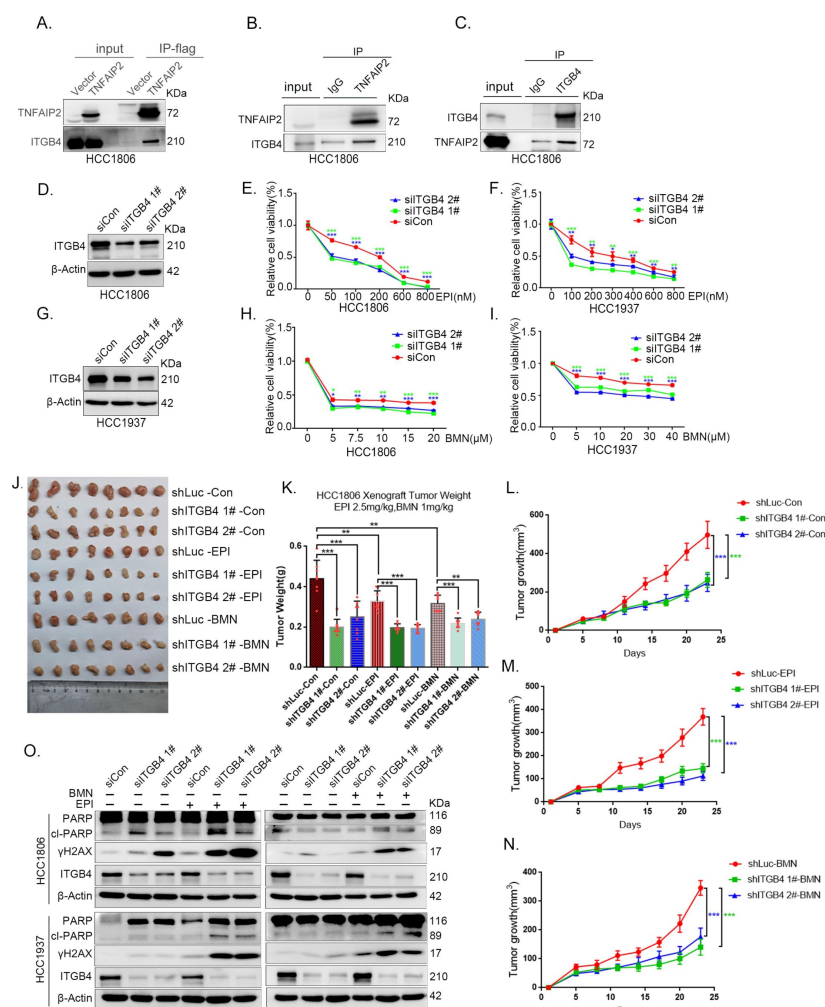


Figure 5.

ITGB4 interacts with TNFAIP2 and promotes TNBC drug resistance and DNA damage repair

(A) TNFAIP2 interacts with ITGB4. HCC1806 cells with stable TNFAIP2 overexpression were collected from flag-tagged TNFAIP2 cell lysates for immunoprecipitation assays using Flag-M2 beads, and ITGB4 was detected by WB.

(B) Endogenous TNFAIP2 interacts with ITGB4 in HCC1806 cells. Endogenous TNFAIP2 protein was immunoprecipitated using an anti-TNFAIP2 antibody. IgG served as the negative control. Endogenous ITGB4 was detected by WB.

(C) Endogenous ITGB4 interacts with TNFAIP2 in HCC1806 cells. Endogenous ITGB4 protein was immunoprecipitated using an anti-ITGB4 antibody. IgG served as the negative control. Endogenous TNFAIP2 was detected by WB.

(D-I) ITGB4 knockdown in HCC1806 and HCC1937 cells significantly decreased cell viability in the presence of EPI (0-800 nM) and BMN (0-40 μ M), as measured by the SRB assay. ITGB4 protein expression was detected by WB.

(J-N) ITGB4 depletion promotes HCC1806 breast cancer cell sensitivity to EPI and BMN treatment *in vivo*. HCC1806 cells with stable ITGB4 knockdown were transplanted into the fat pad of 7-week-old female nude mice. When the average tumor size reached approximately 50 mm⁽³⁾ after inoculation, the mice in each group were randomly divided into two subgroups (n=4/group) to receive EPI (2.5 mg/kg), BMN (1 mg/kg) or vehicle control for 22 days(J). Tumor masses were collected and weighed at the end of the experiments(K), and tumor size was measured twice a week (L-N).*: $P < 0.05$, **: $P < 0.01$, *t*-test.

(O) ITGB4 knockdown increased DNA damage of EPI and BMN. HCC1806 and HCC1937 cells with ITGB4 knockdown were treated with 400 nM EPI for 24h and 5 μ M BMN for 24h, respectively. ITGB4, γ H2AX, and PARP protein expression was detected by WB.

To explore the function of ITGB4 in TNBC drug resistance, we knocked down ITGB4 in HCC1806 and HCC1937 cells. As shown in [Figure 5D-I](#), knockdown of ITGB4 significantly decreased cell viability in the presence of EPI and BMN in both cell lines. Knockdown of ITGB4 also suppressed HCC1806 xenograft growth *in vivo*. More importantly, ITGB4 knockdown further decreased tumor volume when mice were treated with EPI and BMN ([Figure. 5J-N](#)). Meanwhile, BMN treatment had no effect on the body weight of mice, but EPI treatment decreased mouse body weight due to its toxicity (Fig. S5H). We then examined the effects of ITGB4 on DNA damage repair and found that ITGB4 promotes DNA damage repair in response to EPI and BMN. ITGB4 knockdown increased γ H2AX and cleaved-PARP protein expression levels when HCC1806 and HCC1937 cells were treated with EPI and BMN ([Fig. 5O](#)). Furthermore, the function of ITGB4 was validated by using two other drugs, DDP and AZD (Fig. S5A-G). These results suggested that ITGB4 increases TNBC drug resistance and promotes DNA damage repair.

ITGB4 activates Rac1 through TNFAIP2 and IQGAP1

It is well known that ITGB4 can activate Rac1⁽²⁷⁾ and that TNFAIP2 interacts with Rac1 and activates it⁽⁹⁾. To test whether ITGB4 activates Rac1 through TNFAIP2, we measured the levels of GTP-bound Rac1 in ITGB4-overexpressing and ITGB4-knockdown cells. Overexpression of ITGB4 significantly increased the levels of GTP-bound Rac1 in both HCC1806 and HCC1937 cells ([Fig. 6A](#)). In agreement with this observation, knockdown of ITGB4 significantly decreased the levels of GTP-bound Rac1 in both cell lines ([Fig. 6B](#)). Next, we knocked down TNFAIP2 in ITGB4-overexpressing HCC1806 and HCC1937 cells and found that ITGB4-increased Rac1 activity was blocked by TNFAIP2 knockdown ([Fig. 6C-D](#)). Collectively, these results demonstrate that ITGB4 activates Rac1 through TNFAIP2.

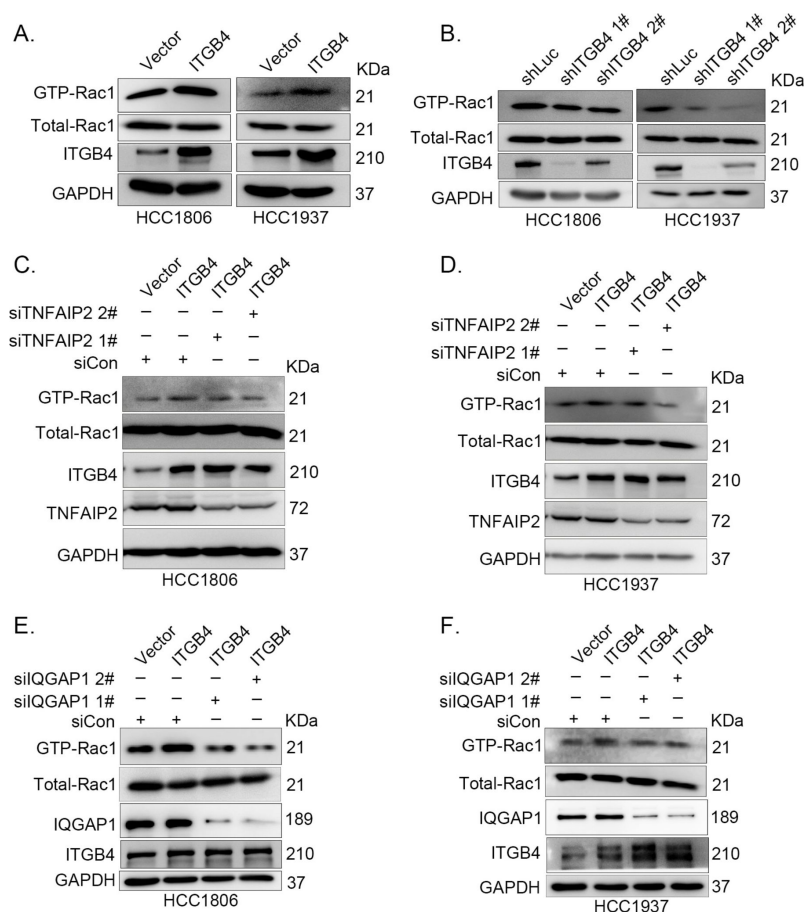


Figure 6.

ITGB4 activates Rac1 through TNFAIP2 and IQGAP1

(A) Overexpression of ITGB4 increased GTP-Rac1 levels in HCC1806 and HCC1937 cells. GTP-Rac1 were assessed using PAK-PBD beads.

(B) Knockdown of ITGB4 by shRNA decreased GTP-Rac1 levels in HCC1806 and HCC1937 cells.

(C-D) ITGB4 activates Rac1 through TNFAIP2. HCC1806 (C) and HCC1937 (D) cells with stable ITGB4 overexpression were transfected with TNFAIP2 or control siRNA.

(E-F) ITGB4 activates Rac1 through IQGAP1. HCC1806 (E) and HCC1937 (F) cells with stable ITGB4 overexpression were transfected with IQGAP1 or control siRNA.

It has been reported that Rac1 activity is promoted by IQGAP1⁽³⁴⁾ and that TNFAIP2 activates Rac1 through IQGAP1 (Fig. 4P). We wondered whether ITGB4 activates Rac1 through IQGAP1; therefore, we knocked down IQGAP1 in HCC1806 and HCC1937 cells with stable overexpression of ITGB4 and found that the ITGB4-induced increase in Rac1 activity was abolished by IQGAP1 knockdown (Fig. 6E-F). These results suggest that ITGB4 activates Rac1 through TNFAIP2 and IQGAP1.

ITGB4 promotes TNBC drug resistance via TNFAIP2/IQGAP1/Rac1

Since ITGB4, TNFAIP2, and IQGAP1 promote drug resistance by promoting DNA damage repair in TNBC, we wondered whether ITGB4 promoted drug resistance through the TNFAIP2/IQGAP1/Rac1 axis. We knocked down TNFAIP2, IQGAP1, and Rac1 in ITGB4-overexpressing cells and found that blocking the TNFAIP2/IQGAP1/Rac1 axis increased the sensitivity of ITGB4-overexpressing HCC1806 (Fig. 7A-I) and HCC1937 cells to EPI and BMN (Fig. S6O-W). We also found that γ H2AX and cleaved-PARP levels were upregulated in TNFAIP2/IQGAP1/Rac1 knockdown HCC1806 and HCC1937 cells stably expressing ITGB4 in the presence of EPI and BMN (Fig. 7J-L, Fig. S6X-Z). DDP and AZD treatment generated similar results (Fig. S6A-N). Together, these results suggest that ITGB4 promotes DNA damage repair and drug resistance via the TNFAIP2/IQGAP1/Rac1 axis.

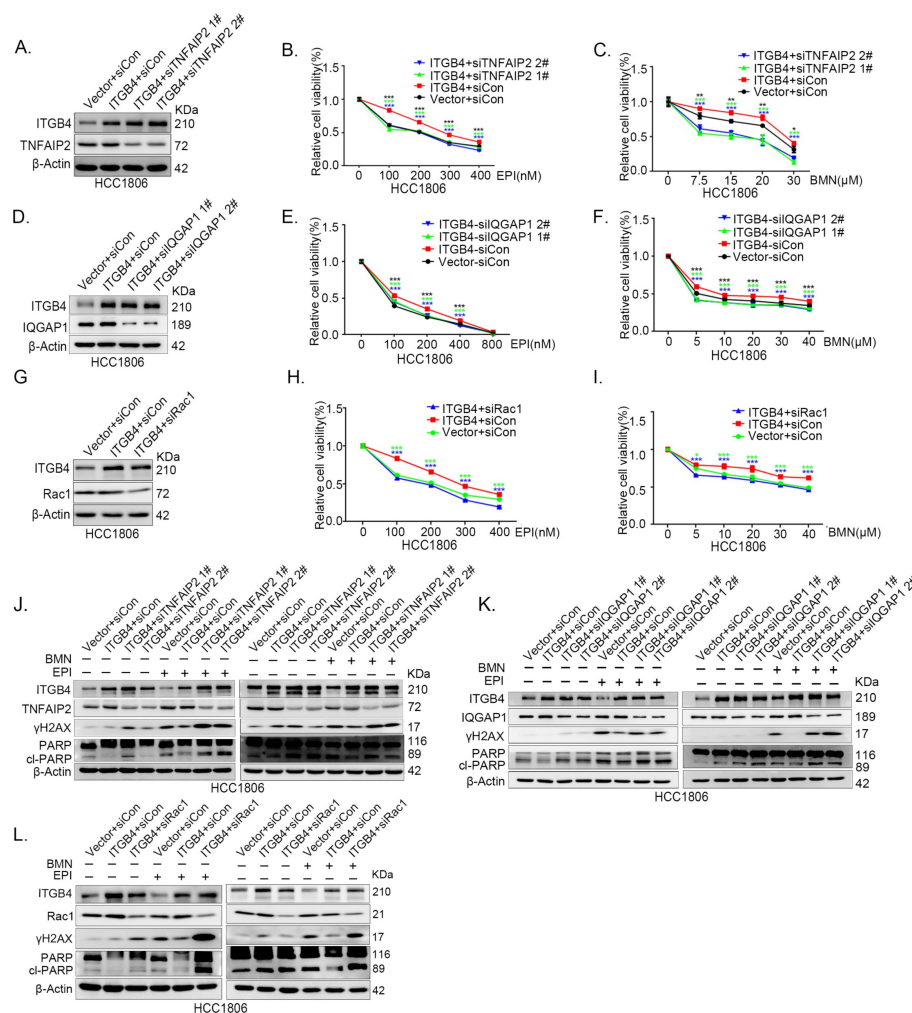


Figure 7.

ITGB4 promotes TNBC drug resistance via TNFAIP2/IQGAP1/Rac1

(A-C) ITGB4 promotes TNBC drug resistance through TNFAIP2. TNFAIP2 knockdown abolished ITGB4-induced resistance to EPI and BMN. HCC1806 cells with stable ITGB4 overexpression were transfected with TNFAIP2 or control siRNA, followed by treatment with EPI (0-400 nM) and BMN (0-30 μM) for 48 or 72 h, respectively. Cell viability was measured by the SRB assay. Protein expression levels were analyzed by WB.

(D-F) ITGB4 promotes TNBC drug resistance through IQGAP1. HCC1806 cells with stable ITGB4 overexpression were transfected with IQGAP1 or control siRNA, followed by treatment with EPI (0-800 nM) and BMN (0-40 μM) for 48 or 72 h, respectively. Cell viability was measured by the SRB assay. Protein expression levels were analyzed by WB.

(G-I) ITGB4 promotes TNBC drug resistance through Rac1. HCC1806 cells with stable ITGB4 overexpression were transfected with Rac1 or control siRNA, followed by treatment with EPI (0-400 nM) and BMN (0-40 μM) for 48 or 72 h, respectively. Cell viability was measured by the SRB assay. Protein expression levels were analyzed by WB.

(J) ITGB4 promotes DNA damage repair through TNFAIP2. HCC1806 cells with stable ITGB4 overexpression were transfected with TNFAIP2 or control siRNA, followed by treatment with EPI (400 nM) and BMN (5 μM) for 24h. Protein expression levels were analyzed by WB.

(K) ITGB4 promotes DNA damage repair through IQGAP1. HCC1806 cells with stable ITGB4 overexpression were transfected with IQGAP1 or control siRNA, followed by treatment with EPI (400 nM) and BMN (5 μM) for 24h. Protein expression levels were analyzed by WB.

(L) ITGB4 promotes DNA damage repair through Rac1. HCC1806 cells with stable ITGB4 overexpression were transfected with Rac1 or control siRNA, followed by treatment with EPI (400 nM) and BMN (5 μM) for 24h. Protein expression levels were analyzed by WB.

TNFAIP2 expression levels positively correlated with ITGB4 in TNBC tissues

To test whether ITGB4 and TNFAIP2 are coexpressed in TNBC, we collected 135 TNBC specimens for IHC (the IQGAP1 antibody did not work for IHC). Clinical pathological parameters, including patient age, tumor size, lymph node status at the time of diagnosis, and follow-up status, including adjuvant treatment and tumor recurrence, were retrospectively obtained from the Department of Pathology, Henan Provincial People's Hospital, Zhengzhou University, China. We performed IHC analyses on two breast cancer tissue chips containing a total of 135 patients with TNBC (Fig. 8A-D). TNFAIP2 and ITGB4 protein expression levels were significantly positively correlated (Table 1).

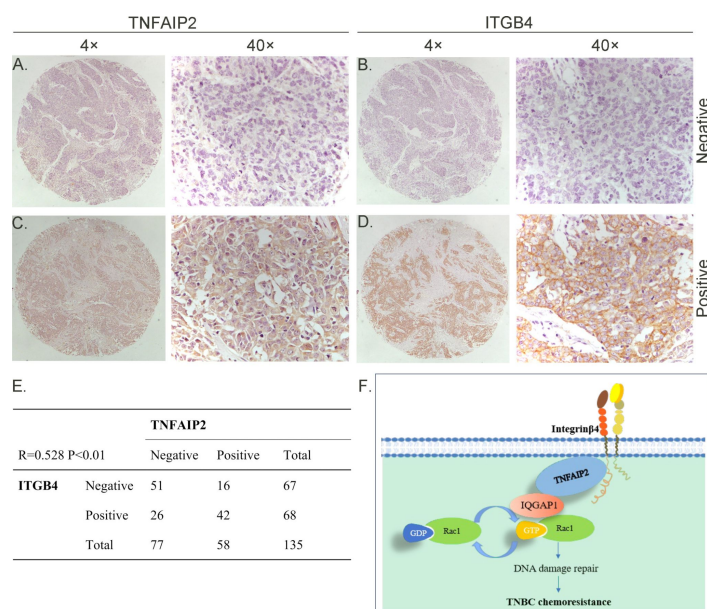


Figure 8.

TNFAIP2 expression levels positively correlated with ITGB4 in TNBC tissues

Representative IHC images of TNFAIP2 and ITGB4 protein expression in breast cancer tissues are shown. The final expression assessment was performed by combining the two scores (0–2=low, 6–7=high). A and B indicate low scores, C and D indicate high scores, and Figure E is the work model of this paper.

Discussion

Chemotherapies, including EPI and DDP, are the main choice for TNBC patients. Unfortunately, TNBC frequently develops resistance to chemotherapy⁽³⁹⁾. Currently, PARP inhibitors are effective for TNBC with BRCA1/2 mutation or homologous recombination deficiency (HRD)^{(40)–(42)}. PARP inhibitors can cause DNA damage repair defects and have synergistic lethal effects with HRD. Meanwhile, chemotherapy and PARP inhibitor resistance is also a major problem in the clinic.

In this study, we first found that TNFAIP2 promotes TNBC drug resistance and DNA damage repair through Rac1. Next, we found that TNFAIP2 interacts with IQGAP1 and ITGB4. We verified that ITGB4 promotes TNBC drug resistance and DNA damage repair through the TNFAIP2/IQGAP1/Rac1 axis.

Interestingly, we discovered for the first time that ITGB4 and TNFAIP2 promote Rac1 activity through IQGAP1. Our study reveals that ITGB4 promotes TNBC resistance through TNFAIP2-, IQGAP1-, and Rac1-mediated DNA damage repair (Figure 7). This study provides new targets for reversing TNBC resistance.

ITGB4 is well known to promote breast cancer stemness and can be activated by laminin 5⁽⁴³⁾. In addition, ITGB4 is generally in partner with ITGA6, which is another marker of

breast cancer stem cells⁽⁴⁴⁾ and drug resistance⁽⁴³⁾. Therefore, whether ITGA6 has similar functions needs further study. It was reported that ITGB4 activates Rac1⁽⁴⁵⁾, but the mechanism is unclear. For the first time, we revealed that ITGB4 activates Rac1 through TNFAIP2 and IQGAP1. More importantly, ITGB4 promotes drug resistance through the TNFAIP2/IQGAP1/Rac1 axis.

TNFAIP2 plays important roles in different cellular and physiological processes, including cell proliferation, adhesion, migration, membrane TNT formation, angiogenesis, inflammation and tumorigenesis⁽¹⁴⁾. We previously found that TNFAIP2 was regulated by KLF5 and interacted with the small GTPases Rac1 and Cdc42, thereby regulating the actin cytoskeleton and cell morphology in breast cancer cells⁽⁹⁾. However, the detailed mechanism is not clearly understood. In this study, we found that IQGAP1 mediates this process. IQGAP1 is a crucial regulator of cancer development by scaffolding and facilitating different oncogenic pathways, especially Rac1/Cdc42, thus affecting proliferation, adhesion, migration, invasion, and metastasis⁽⁴⁶⁾. In addition, IQGAP1 is increased during the differentiation of ovarian cancer stem cells and promotes aggressive behaviors⁽⁴⁷⁾. In our study, we found that TNFAIP2 interacts with IQGAP1 and thus activates Rac1 to induce chemotherapy and PARP inhibitor drug resistance.

Furthermore, TNFAIP2 was reported to induce epithelial-to-mesenchymal transition and confer platinum resistance in urothelial cancer cells⁽¹²⁾, and in embryonic stem cell (ESC) differentiation, TNFAIP2 was found to be important in controlling lipid metabolism, which supports the ESC differentiation process and planarian organ maintenance⁽⁴⁸⁾. Another study found that TNFAIP2 overexpression enhanced TNT-mediated autophagosome and lysosome exchange, preventing advanced glycation end product (AGE)-induced autophagy and lysosome dysfunction and apoptosis⁽⁴⁹⁾. In cancer treatment, TNFAIP2 was chosen as one of the six signature genes predicting chemotherapeutic and immunotherapeutic efficacies, with high-senescore patients benefiting from immunotherapy and low-senescore patients responsive to chemotherapy⁽⁵⁰⁾.

These reports provide a possible explanation for previous studies showing that ITGB4 is important in EMT and cancer stemness. According to our results that there is an interaction between ITGB4 and TNFAIP2, ITGB4 might regulate EMT and stemness through TNFAIP2. TNFAIP2 is one of the important factors induced by tumor necrosis factor alpha (TNFα). Interestingly, TNFα release could be induced by therapeutic drugs from multiple tumor cell lines. The acquisition of docetaxel resistance was accompanied by increased constitutive production of TNFα⁽⁵¹⁾. In addition, TNFα is a key tumor-promoting effector molecule secreted by tumor-associated macrophages. *In vitro* neutralizing TNFα was observed to inhibit tumor progression and improve the curative effect of bevacizumab⁽⁵²⁾. Therefore, the mechanism by which TNFα plays a role in chemotherapeutic resistance in breast cancer should be further investigated. TNFα neutralizing provides novel opinions elucidating drug resistance.

For future studies, it will be important to develop *Tnfaip2* knockout mice to investigate the exact role of TNFAIP2 physiologically. According to recent studies and our findings, agents targeting the interaction among ITGB4/TNFAIP2/IQGAP1 would be a promising trend for developing drugs to overcome the resistance phenomenon.

In summary, ITGB4 and TNFAIP2 play important roles in breast cancer chemoresistance. TNFAIP2 activates Rac1 to promote chemoresistance through IQGAP1. In addition, ITGB4 activates Rac1 through TNFAIP2 and IQGAP1 and confer DNA damage drug resistance in TNBC (Figure 8 E). and HCC1937 cells were cultured in RPMI. These results indicate that the ITGB4/TNFAIP2/IQGAP1/Rac1 axis provides potential therapeutic targets to overcome DNA damage drug resistance in TNBC.

Data availability statement

The authors confirm that the data supporting the findings of this study are available within the article and its supplementary materials.

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Author contributions

H.F., W.R., Q.C., H.L. and C.Y. conducted different parts of the experiments. W.L., X.W., X.L. and Y.S. provided some advice about this project. C.C. and J.F. designed the study. C.C., J.F., and Y.S. supervised the study. H.F. wrote the original draft and W.R., Q.C., H.L., C.C. and J.F. reviewed and edited the draft.

Declaration of interests

The authors declare no conflicts of interest.

Table 1. The correlation of TNFAIP2 and ITGB4 protein expression levels in human TNBC specimens.

STAR Methods

Cell lines and reagents

All cell lines used in this study, including HCC1806, HCC1937, and HEK293T cells, were purchased from ATCC (American Type Culture Collection, Manassas, VA, USA) and validated by STR (short tandem repeat) analysis. HCC1806 and HCC1937 cells were cultured in RPMI 1640 medium supplemented with 5% FBS. HEK293T cells were cultured in DMEM (Thermo Fisher, Grand Island, USA) with 5% FBS at 37°C with 5% CO₂. Epirubicin(EPI) (Cat#HY-13624A), cisplatin(DDP)(Cat#HY-17394), talazoparib (BMN) (Cat#HY-16106), and olaparib (AZD) (Cat#HY-10162) were purchased from MCE (New Jersey, USA).

Plasmid construction and stable TNFAIP2 and ITGB4 overexpression

We constructed the full-length *TNFAIP2/ITGB4* gene and then subcloned them into the pCDH lentiviral vector. The packaging plasmids (including pMDLg/pRRE, pRSV-Rev, and pCMV-VSV-

G) and pCDH-TNFAIP2/ITGB4 expression plasmid were cotransfected into HEK293T cells (2×10^6 in 10 cm plate) to produce lentivirus. Following transfection for 48 h, the lentivirus was collected and used to infect HCC1806 and HCC1937 cells. Forty-eight hours later, puromycin (2 μ g/ml) was used to screen the cell populations.

Stable knockdown of TNFAIP2 and ITGB4

The pSIH1-H1-puro shRNA vector was used to express TNFAIP2, ITGB4 and luciferase(LUC)shRNAs. TNFAIP2shRNA#1, 5'-GACUUGGGCUCACAGAUAA-3'; TNFAIP2shRNA#2, 5'-GAUUGAGGUGGCCACUUUAU-3'; ITGB4shRNA#1, 5'-ACGACAGCTTCCTTATGTA-3'; ITGB4shRNA#2, 5'-CAGCGACTACACTATTG GA-3'; LuciferaseshRNA, 5'-CUUACGUGAGUACUUCGA-3'; HCC1806 and HCC1937 cells were infected with lentivirus. Stable populations were selected using 1 to 2 mg/mL puromycin. The knockdown effect was evaluated by Western blotting.

RNA interference

The siRNA target sequences used in this study are as follows: TNFAIP2siRNA#1, 5'-GACUUGGGCUCACAGAUAA-3'; TNFAIP2siRNA#2, 5'-GAUUGAGGUGGCCACUUUAU-3'; ITGB4siRNA#1, 5'-ACGACAGCTTCCTTAT GTA-3'; ITGB4siRNA#2, 5'-CAGCGACTACACTATTGGA-3'; Rac1siRNA, 5'-CG GCACCACUGUCCCAACA-3'; IQGAP1siRNA#1, 5'-GCAGGTGGATTACTATAAA-3'; IQGAP1siRNA#2, 5'-CUAGUGAAACUGCAACAGA-3'. All siRNAs were synthesized by RiboBio (RiboBio, China) and transfected at a final concentration of 50 nM.

Antibodies and Western blotting (WB)

The WB procedure has been described in our previous study⁽⁵³⁾. Anti-TNFAIP2 (sc-28318), anti-ITGB4 (sc-9090) and anti-GAPDH (sc-25778) antibodies were purchased from Santa Cruz Biotechnology (Santa Cruz, CA, USA). The anti-PARP (#9542) antibody was purchased from CST. Anti-Rac1 (05-389) and anti- γ H2AX (3475627) antibodies were purchased from Millipore (Billerica, MA, USA). Anti- β -actin (A5441) and anti-Tubulin (T5168) antibodies were purchased from Sigma-Aldrich (St Louis, MO, USA). The anti-IQGAP1 (ab86064) antibody was purchased from Abcam.

Immunopurification and silver staining

Immunoprecipitation and silver staining lysates from HCC1806 cells stably expressing Flag-TNFAIP2 were prepared by incubating the cells in lysis buffer containing a protease inhibitor cocktail (MCE). Cell lysates were obtained from approximately 2.5×10^8 cells, and after binding with anti-Flag M2 affinity gel (Sigma) for 2 h as recommended by the manufacturer, the affinity gel was washed with cold lysis plus 0.2% NP-40. FLAG peptide (Sigma) was applied to elute the Flag-labeled protein complex as described by the vendor. The elutes were collected and visualized on NuPAGE 4%-12% Bis-Tris gels (Invitrogen, CA, USA) followed by silver staining with a silver staining kit (Pierce, Illinois, USA). The distinct protein bands were retrieved and analyzed by LC-Mass.

Immunoprecipitation and GST pull-down

For exogenous interaction between ITGB4 and Flag-TNFAIP2, cell lysates were directly incubated with anti-Flag M2 affinity gel (A2220; Sigm) overnight at 4°C. For endogenous protein interaction, cell lysates were first incubated with anti-TNFAIP2/ITGB4/IQGAP1 antibodies or mouse IgG/rabbit IgG (sc-2028; Santa Cruz Biotechnology, CA, USA) and then

incubated with Protein A/G plus agarose beads (sc-2003; Santa Cruz Biotechnology) for the GST pulldown assay, cell lysates were directly incubated with Glutathione Sepharose 4B (52-2303-00; GE Healthcare) overnight at 4°C. The precipitates were washed four times with 1 ml of lysis buffer, boiled for 10 minutes with 1×SDS sample buffer, and subjected to WB analysis.

Cell viability assays

Cell viability was measured by SRB assays as described in our previous study⁽⁵⁴⁾. Cell viability was measured by SRB assays. Briefly, cells were seeded in 96-well plates. Then, the cells were cultured for the indicated time and fixed with 10% trichloroacetic acid at room temperature for 30 min, followed by incubation with 0.4% SRB (w/v) solution in 1% acetic acid for 20 min at room temperature. Finally, SRB was dissolved in 10 mM unbuffered Tris base, and the absorbance was measured at a wavelength of 530 nm on a plate reader (Bio Tek, Vermont, USA).

Rac1 activation assays

Rac1 activation was examined using the Cdc42 Activation Assay Biochem Kit (BK034, Cytoskeleton, Denver, USA) following the manufacturer's instructions. Cells were harvested with cell lysis buffer, and 1 mg of protein lysate in a 1 ml total volume at 4°C was immediately precipitated with 10 µg of PAK-PBD beads for 60 min with rotation. After washing three times with wash buffer, agarose beads were resuspended in 30 µl of 2×SDS sample buffer and boiled for 5 min. Rac1-GTP was examined by WB with an anti-Rac1 antibody.

Xenograft experiments

We purchased 6- to 7-week-old female BALB/c nude mice from SLACCAS (Changsha, China). Animal feeding and experiments were approved by the animal ethics committee of Kunming Institute of Zoology, Chinese Academy of Sciences. HCC1806-shLuc, HCC1806-shTNFAIP2, or HCC1806-shITGB4 cells (1×10^6) in Matrigel (BD Biosciences, NY, USA)) were implanted into the mammary fat pads of the mice. When the tumor volume reached approximately 50 mm⁽³⁾, the nude mice were randomly assigned to the control and treatment groups (n = 4/group). EPI, BMN, and DDP were dissolved in ddH₂O. The control group was given vehicle alone, and the treatment group received EPI (2.5 mg/kg), BMN (1 mg/kg), and DDP (2.5 mg/kg) alone via intraperitoneal injection every three days for 18 or 27 days. The tumor volume was calculated as follows: tumor volume was calculated by the formula: $(\pi \times \text{length} \times \text{width}^2)/6$.

Immunohistochemical staining

Paraffin-embedded clinical TNBC specimens were obtained from the First Affiliated Hospital, Zhengzhou University, Zhengzhou, China. Informed consent was obtained from all subjects. Two tissue microarrays containing 135 TNBC breast cancer tissues were constructed. For the immunohistochemistry (IHC) assay, the slides were deparaffinized, rehydrated, and pressure cooker heated for 2.5 min in EDTA for antigen retrieval. Endogenous peroxidase activity was inactivated by adding an endogenous peroxidase blocker (OriGene, China) for 15 min at room temperature. Slides were incubated overnight at 4°C with anti-TNFAIP2 (1:200) or anti-ITGB4 (1:500). After 12 h, the slides were washed three times with PBS and incubated with secondary antibodies (hypersensitive enzyme-labeled goat anti-mouse/rabbit IgG polymer (OriGene, China) at room temperature for 20 min, DAB concentrate chromogenic solution (1:200 dilution of concentrated DAB

chromogenic solution), counterstained with 0.5% hematoxylin, dehydrated with graded concentrations of ethanol for 3 min each (70%–80%–90%–100%), and finally stained with dimethyl benzene immunostained slides were evaluated by light microscopy. The IHC signal was scored using the 'Allred Score' method.

Statistical analysis

All graphs were created using GraphPad Prism software version 5.0. Comparisons between two independent groups were assessed by two-tailed Student's t-test. One-way analysis of variance with least significant differences was used for multiple group comparisons. *P*-values of <0.05, 0.01 or 0.001 were considered to indicate a statistically significant result, comparisons significant at the 0.05 level are indicated by *, at the 0.01 level are indicated by **, or at the 0.001 level are indicated by ***.

Supplemental information titles and legends

FigureS1. TNFAIP2 promotes TNBC DNA damage drug resistance

(A-E) Stable TNFAIP2 overexpression in HCC1806 and HCC1937 cells significantly increased cell viability in the presence of DDP (0–40 μ M) or AZD (0–40 μ M) treatment for 48h, as measured by the SRB assay. TNFAIP2 protein expression was detected by WB.

(F-J) Stable TNFAIP2 knockdown in HCC1806 and HCC1937 cells significantly decreased cell viability in the presence of DDP (0–40 μ M) or AZD (0–40 μ M) treatment for 48h, as measured by the SRB assay. TNFAIP2 protein expression was detected by WB.

(K) TNFAIP2 promoted DNA damage repair in the presence of DDP and AZD. HCC1806 and HCC1937 cells stably overexpressing TNFAIP2 were treated with 20 μ M DDP for 24 h or 48h and 10 μ M AZD for 24h, respectively. TNFAIP2, γ H2AX, and PARP protein expression was detected by WB.

(L) TNFAIP2 knockdown increased DNA damage in the presence of DDP and AZD. Stable TNFAIP2 knockdown cells were treated with 2.5 or 20 μ M DDP for 24h and 2.5 μ M AZD for 24h. TNFAIP2, γ H2AX, and PARP protein expression was detected by WB.

FigureS2. TNFAIP2 confers TNBC drug resistance *in vivo*

(A-F) TNFAIP2 knockdown increased the sensitivity of HCC1806 breast cancer cells to DDP *in vivo*. HCC1806 cells with stable TNFAIP2 knockdown were transplanted into the fat pad of 7-week-old female nude mice. When the average tumor size reached approximately 50 mm³ after inoculation, mice in each group were randomly divided into two subgroups (n=4/group) to receive DDP (2.5 mg/kg) or vehicle control for 23 days (A). Tumor size was measured twice a week (B), tumor masses were collected and weighed at the end of the experiments (C), and mouse masses were collected and weighed at the beginning or end of the experiments (D-F). *: *P*<0.05, **: *P*<0.01, *t*-test.

FigureS3. TNFAIP2 promotes TNBC drug resistance and DNA damage repair via Rac1

(A-F) Rac1 knockdown abolished TNFAIP2-induced TNBC resistance to DDP and AZD. HCC1806 (A-C) and HCC1937 (D-F) cells with stable TNFAIP2 overexpression were transfected with Rac1 or control siRNA, followed by treatment with DDP (0-40 μ M) and AZD (0-40 μ M) for 48 or 72 h, respectively. Cell viability was measured by the SRB assay. Protein expression levels were analyzed by WB.

(G-J) Rac1 depletion abolished TNFAIP2-induced DNA damage decrease in response to DDP and AZD. HCC1806 (G-H) and HCC1937 (I-J) cells with stable TNFAIP2 overexpression were transfected with Rac1 or control siRNA, followed by treatment with DDP (20 μ M) and AZD (10 μ M) for 24h, respectively. Protein expression levels were analyzed by WB.

FigureS4. IQGAP1 mediates Rac1 activation by TNFAIP2 and promotes TNBC drug resistance

(A) TNFAIP2 interacts with IQGAP1. HCC1806 cells with stable TNFAIP2 overexpression were collected from flag-tagged TNFAIP2 cell lysates for immunoprecipitation assays using Flag-M2 beads, and IQGAP1 was detected by WB.

(B) Mapping the domains of TNFAIP2 that interact with IQGAP1. Flag-tagged full-length or truncated TNFAIP2 was transfected into HEK293T cells with no-tagged full-length IQGAP1. Cell lysates were collected for immunoprecipitation using Flag-M2 beads, and IQGAP1 was detected by WB.

(C-E) IQGAP1 knockdown abolished TNFAIP2-conferred resistance to EPI and BMN. HCC1937 cells with stable TNFAIP2 overexpression were transfected with IQGAP1 or control siRNA, followed by treatment with EPI (0-1600 nM) and BMN (0-40 μ M) for 48 or 72 h, respectively. Cell viability was measured by the SRB assay, and IQGAP1 was detected by WB.

FigureS5. ITGB4 interacts with TNFAIP2 and promotes TNBC drug resistance and DNA damage repair

(A-F) ITGB4 knockdown in HCC1806 and HCC1937 cells significantly decreased cell viability in the presence of DDP (0-40 μ M) and AZD (0-40 μ M), as measured by the SRB assay. ITGB4 protein expression was detected by WB.

(G) ITGB4 knockdown increased DNA damage of DDP and AZD. HCC1806 and HCC1937 cells with ITGB4 knockdown were treated with 5 μ M or 7.5 μ M DDP for 24h and 15 μ M or 20 μ M AZD for 24h, respectively. ITGB4, γ H2AX, and PARP protein expression was detected by WB.

(H) Mouse masses were collected and weighed at the end of the experiments.

(I) The model of full-length or truncated TNFAIP2.

(J) Mapping the domains of TNFAIP2 that interact with ITGB4. GST-tagged Full-length ITGB4 was transfected into HEK293T cells with flag-tagged full-length or truncated TNFAIP2. TNFAIP2 protein was immunoprecipitated using Flag-M2 beads, and ITGB4 was detected by WB.

(K) Mapping the domains of TNFAIP2 that interact with ITGB4. No-tagged full-length ITGB4 was transfected into HEK293T cells with GST-tagged full-length or truncated TNFAIP2. Cell lysates were collected for the GST pull-down assay, and ITGB4 was detected by WB.

(L) The model of full-length or truncated ITGB4.

(M) Mapping the domains of ITGB4 that interact with TNFAIP2. Flag-tagged full-length TNFAIP2 was transfected into HEK293T cells with no-tagged full-length or truncated ITGB4. Cell lysates were collected for immunoprecipitation using Flag-M2 beads, and ITGB4 was detected by WB.

FigureS6. ITGB4 promotes TNBC drug resistance via TNFAIP2/IQGAP1/Rac1

(A-F) ITGB4 promotes TNBC drug resistance through TNFAIP2. TNFAIP2 knockdown abolished ITGB4-induced resistance to DDP and AZD. HCC1806 (A-C) and HCC1937 (D-F) cells with stable ITGB4 overexpression were transfected with TNFAIP2 or control siRNA, followed by treatment with DDP (0-30 μ M) and AZD (0-30 μ M) for 48 or 72h, respectively. Cell viability was measured by the SRB assay. Protein expression levels were analyzed by WB.

(G-L) ITGB4 promotes TNBC drug resistance through Rac1. Rac1 knockdown abolished ITGB4-induced resistance to DDP and AZD. HCC1806 (G-I) and HCC1937 (J-L) cells with stable ITGB4 overexpression were transfected with Rac1 or control siRNA, followed by treatment with DDP (0-20 μ M) and AZD (0-40 μ M) for 48 or 72h, respectively. Cell viability was measured by the SRB assay. Protein expression levels were analyzed by WB.

(M) ITGB4 promotes DNA damage repair through TNFAIP2. HCC1806 and HCC1937 cells with stable ITGB4 overexpression were transfected with TNFAIP2 or control siRNA, followed by treatment with DDP (7.5 μ M or 10 μ M) and AZD (20 μ M or 30 μ M) for 24h. Protein expression levels were analyzed by WB.

(N) ITGB4 promotes DNA damage repair through Rac1. HCC1806 and HCC1937 cells with stable ITGB4 overexpression were transfected with Rac1 or control siRNA, followed by treatment with DDP (7.5 μ M) and AZD (20 μ M or 30 μ M) for 24h. Protein expression levels were analyzed by WB.

(O-Q) ITGB4 promotes TNBC drug resistance through TNFAIP2. TNFAIP2 knockdown abolished ITGB4-induced resistance to EPI and BMN. HCC1937 cells with stable ITGB4 overexpression were transfected with TNFAIP2 or control siRNA, followed by treatment with EPI (0-800 nM) and BMN (0-30 μ M) for 48 or 72 h, respectively. Cell viability was measured by the SRB assay. Protein expression levels were analyzed by WB.

(R-T) ITGB4 promotes TNBC drug resistance through IQGAP1. HCC1937 cells with stable ITGB4 overexpression were transfected with IQGAP1 or control siRNA, followed by treatment with EPI (0-800 nM) and BMN (0-40 μ M) for 48 or 72 h, respectively. Cell viability was measured by the SRB assay. Protein expression levels were analyzed by WB.

(U-W) ITGB4 promotes TNBC drug resistance through Rac1. HCC1937 cells with stable ITGB4 overexpression were transfected with Rac1 or control siRNA, followed by treatment with EPI (0-800 nM) and BMN (0-30 μ M) for 48 or 72 h, respectively. Cell viability was measured by the SRB assay. Protein expression levels were analyzed by WB.

(X) ITGB4 promotes DNA damage repair through TNFAIP2. HCC1937 cells with stable ITGB4 overexpression were transfected with TNFAIP2 or control siRNA, followed by treatment with EPI (800 nM) and BMN (10 μ M) for 24h. Protein expression levels were analyzed by WB.

(Y) ITGB4 promotes DNA damage repair through IQGAP1. HCC1937 cells with stable ITGB4 overexpression were transfected with IQGAP1 or control siRNA, followed by treatment with EPI (800 nM) and BMN (10 μ M) for 24h. Protein expression levels were analyzed by WB.

(Z) ITGB4 promotes DNA damage repair through Rac1. HCC1937 cells with stable ITGB4 overexpression were transfected with Rac1 or control siRNA, followed by treatment with EPI (800 nM) and BMN (10 μ M) for 24h. Protein expression levels were analyzed by WB.

References

1. Bray F. , Ferlay J. , Soerjomataram I. , Siegel R.L. , Torre L.A. , Jemal A. (2018) **Global cancer statistics 2018: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries** *CA: a cancer journal for clinicians* **68**:394–424
<https://doi.org/10.3322/caac.21492>
2. Zhao S. , Ma D. , Xiao Y. , Li X.M. , Ma J.L. , Zhang H. , Xu X.L. , Lv H. , Jiang W.H. , Yang W.T. , et al. (2020) **Zhao, S., Ma, D., Xiao, Y., Li, X.M., Ma, J.L., Zhang, H., Xu, X.L., Lv, H., Jiang, W.H., Yang, W.T., et al. (2020). Molecular Subtyping of Triple-Negative Breast Cancers by Immunohistochemistry: Molecular Basis and Clinical Relevance. 25, e1481–e1491. 10.1634/theoncologist.2019-0982. Molecular Subtyping of Triple-Negative Breast Cancers by Immunohistochemistry: Molecular Basis and Clinical Relevance 25**
<https://doi.org/10.1634/theoncologist.2019-0982>
3. Sotiriou C. , Pusztai L. (2009) **Gene-expression signatures in breast cancer** *The New England journal of medicine* **360**:790–800
<https://doi.org/10.1056/NEJMra0801289>
4. Bai X. , Ni J. , Beretov J. , Graham P. , Li Y. (2021) **Triple-negative breast cancer therapeutic resistance: Where is the Achilles' heel?** *Cancer letters* **497**:100–111
<https://doi.org/10.1016/j.canlet.2020.10.016>
5. Jamdade V.S. , Sethi N. , Mundhe N.A. , Kumar P. , Lahkar M. , Sinha N. (2015) **Therapeutic targets of triple-negative breast cancer: a review** *British journal of pharmacology* **172**:4228–4237
<https://doi.org/10.1111/bph.13211>
6. Hill D.P. , Harper A. , Malcolm J. , McAndrews M.S. , Mockus S.M. , Patterson S.E. , Reynolds T. , Baker E.J. , Bult C.J. , Chesler E.J. , Blake J.A. (2019) **Cisplatin-resistant triple-negative breast cancer subtypes: multiple mechanisms of resistance** *BMC cancer* **19**
<https://doi.org/10.1186/s12885-019-6278-9>

7. Longley D.B. , Johnston P.G. (2005) **Molecular mechanisms of drug resistance** *The Journal of pathology* **205**:275–292
<https://doi.org/10.1002/path.1706>
8. Rincón R. , Zazo S. , Chamizo C. , Manso R. , González-Alonso P. , Martín-Aparicio E. , Cristóbal I. , Cañadas C. , Perona R. , Lluch A. , et al. (2016) **c-Jun N-Terminal Kinase Inactivation by Mitogen-Activated Protein Kinase Phosphatase 1 Determines Resistance to Taxanes and Anthracyclines in Breast Cancer** *Molecular cancer therapeutics* **15**:2780–2790
<https://doi.org/10.1158/1535-7163.mct-15-0920>
9. Jia L. , Zhou Z. , Liang H. , Wu J. , Shi P. , Li F. , Wang Z. , Wang C. , Chen W. , Zhang H. , et al. (2016) **KLF5 promotes breast cancer proliferation, migration and invasion in part by upregulating the transcription of TNFAIP2** *Oncogene* **35**:2040–2051
<https://doi.org/10.1038/onc.2015.263>
10. Chen L.C. , Chen C.C. , Liang Y. , Tsang N.M. , Chang Y.S. , Hsueh C. (2011) **A novel role for TNFAIP2: its correlation with invasion and metastasis in nasopharyngeal carcinoma** *Modern pathology : an official journal of the United States and Canadian Academy of Pathology, Inc* **24**:175–184
<https://doi.org/10.1038/modpathol.2010.193>
11. Cheng Z. , Wang H.Z. , Li X. , Wu Z. , Han Y. , Li Y. , Chen G. , Xie X. , Huang Y. , Du Z. , Zhou Y. (2015) **MicroRNA-184 inhibits cell proliferation and invasion, and specifically targets TNFAIP2 in Glioma** *Journal of experimental & clinical cancer research : CR* **34**
<https://doi.org/10.1186/s13046-015-0142-9>
12. Niwa N. , Tanaka N. (2019) **Niwa, N., and Tanaka, N. (2019). TNFAIP2 expression induces epithelial-to-mesenchymal transition and confers platinum resistance in urothelial cancer cells. 99, 1702-1713. 10.1038/s41374-019-0285-y. TNFAIP2 expression induces epithelial-to-mesenchymal transition and confers platinum resistance in urothelial cancer cells 99:1702–1713**
<https://doi.org/10.1038/s41374-019-0285-y>
13. Xie Y. , Wang B. (2017) **Downregulation of TNFAIP2 suppresses proliferation and metastasis in esophageal squamous cell carcinoma through activation of the Wnt/ β -catenin signaling pathway** *Oncology reports* **37**:2920–2928
<https://doi.org/10.3892/or.2017.5557>
14. Jia L. , Shi Y. , Wen Y. , Li W. , Feng J. , Chen C. (2018) **Jia, L., Shi, Y., Wen, Y., Li, W., Feng, J., and Chen, C. (2018). The roles of TNFAIP2 in cancers and infectious diseases. 22, 5188-5195. 10.1111/jcmm.13822. The roles of TNFAIP2 in cancers and infectious diseases 22:5188–5195**
<https://doi.org/10.1111/jcmm.13822>
15. Didsbury J. , Weber R.F. , Bokoch G.M. , Evans T. , Snyderman R. (1989) **rac, a novel ras-related family of proteins that are botulinum toxin substrates** *The Journal of biological chemistry* **264**:16378–16382
16. Rul W. , Zugasti O. , Roux P. , Peyssonnaud C. , Eychene A. , Franke T.F. , Lenormand P. , Fort P. , Hibner U. (2002) **Activation of ERK, controlled by Rac1 and Cdc42 via Akt, is required for anoikis** *Annals of the New York Academy of Sciences* **973**:145–148
<https://doi.org/10.1111/j.1749-6632.2002.tb04624.x>

17. Liu W. , Han J. , Shi S. , Dai Y. , He J. (2019) **TUFT1 promotes metastasis and chemoresistance in triple negative breast cancer through the TUFT1/Rab5/Rac1 pathway** *Cancer cell international* **19**
<https://doi.org/10.1186/s12935-019-0961-4>
18. Yan Y. , Greer P.M. , Cao P.T. , Kolb R.H. , Cowan K.H. (2012) **RAC1 GTPase plays an important role in γ -irradiation induced G2/M checkpoint activation** *Breast cancer research : BCR* **14**
<https://doi.org/10.1186/bcr3164>
19. Wu M. , Li L. , Hamaker M. , Small D. , Duffield A.S. (2019) **FLT3-ITD cooperates with Rac1 to modulate the sensitivity of leukemic cells to chemotherapeutic agents via regulation of DNA repair pathways** *Haematologica* **104**:2418–2428
<https://doi.org/10.3324/haematol.2018.208843>
20. Hu H. , Juvekar A. , Lyssiotis C.A. , Lien E.C. , Albeck J.G. , Oh D. , Varma G. , Hung Y.P. , Ullas S. , Lauring J. , et al. (2016) **Phosphoinositide 3-Kinase Regulates Glycolysis through Mobilization of Aldolase from the Actin Cytoskeleton** *Cell* **164**:433–446
<https://doi.org/10.1016/j.cell.2015.12.042>
21. Li Q. , Qin T. , Bi Z. , Hong H. , Ding L. , Chen J. , Wu W. , Lin X. , Fu W. , Zheng F. , Yao Y. (2020) **Li, Q., Qin, T., Bi, Z., Hong, H., Ding, L., Chen, J., Wu, W., Lin, X., Fu, W., Zheng, F., and Yao, Y. (2020). Rac1 activates non-oxidative pentose phosphate pathway to induce chemoresistance of breast cancer. 11, 1456. 10.1038/s41467-020-15308-7. Rac1 activates non-oxidative pentose phosphate pathway to induce chemoresistance of breast cancer 11**
<https://doi.org/10.1038/s41467-020-15308-7>
22. Hervieu A. , Heuss S.F. (2020) **Hervieu, A., and Heuss, S.F. (2020). A PI3K- and GTPase-independent Rac1-mTOR mechanism mediates MET-driven anchorage-independent cell growth but not migration. 13. 10.1126/scisignal.aba8627. A PI3K- and GTPase-independent Rac1-mTOR mechanism mediates MET-driven anchorage-independent cell growth but not migration 13**
<https://doi.org/10.1126/scisignal.aba8627>
23. Li Y. , Li P. , Wang N. (2021) **Effect of let-7c on the PI3K/Akt/FoxO signaling pathway in hepatocellular carcinoma** *Oncology letters* **21**
<https://doi.org/10.3892/ol.2020.12357>
24. Feng Y.X. , Zhao J.S. , Li J.J. , Wang T. , Cheng S.Q. , Yuan Y. , Wang F. , Wang X.F. , Xie D. (2010) **Liver cancer: EphrinA2 promotes tumorigenicity through Rac1/Akt/NF-kappaB signaling pathway** *Hepatology (Baltimore, Md.)* **51**:535–544
<https://doi.org/10.1002/hep.23313>
25. Higuchi M. , Onishi K. , Kikuchi C. , Gotoh Y. (2008) **Scaffolding function of PAK in the PDK1-Akt pathway** *Nature cell biology* **10**:1356–1364
<https://doi.org/10.1038/ncb1795>

26. Hu F. , Li N. , Li Z. , Zhang C. , Yue Y. , Liu Q. , Chen L. , Bilan P.J. , Niu W. (2018) **Electrical pulse stimulation induces GLUT4 translocation in a Rac-Akt-dependent manner in C2C12 myotubes** *FEBS letters* **592**:644–654
<https://doi.org/10.1002/1873-3468.12982>
27. Hamill K.J. , Hopkinson S.B. , DeBiase P. , Jones J.C. (2009) **BPAG1e maintains keratinocyte polarity through beta4 integrin-mediated modulation of Rac1 and cofilin activities** *Molecular biology of the cell* **20**:2954–2962
<https://doi.org/10.1091/mbc.e09-01-0051>
28. Wu J. , Zhao R. , Lin J. , Liu B. (2018) **Integrin $\beta 4$ reduces DNA damage-induced p53 activation in colorectal cancer** *Oncology reports* **40**:2183–2192
<https://doi.org/10.3892/or.2018.6628>
29. Bieri B. , Pierce S.E. , Kroeger C. , Stover D.G. , Pattabiraman D.R. , Thiru P. , Liu Donaher J. , Reinhardt F. , Chaffer C.L. , Keckesova Z. , Weinberg R.A. (2017) **Integrin- $\beta 4$ identifies cancer stem cell-enriched populations of partially mesenchymal carcinoma cells** *Proceedings of the National Academy of Sciences of the United States of America* **114**
<https://doi.org/10.1073/pnas.1618298114>
30. Ruan S. , Lin M. , Zhu Y. , Lum L. , Thakur A. , Jin R. , Shao W. , Zhang Y. , Hu Y. , Huang S. , et al. (2020) **Integrin $\beta 4$ -Targeted Cancer Immunotherapies Inhibit Tumor Growth and Decrease Metastasis** *Cancer research* **80**:771–783
<https://doi.org/10.1158/0008-5472.can-19-1145>
31. Kang S. , Jang Y. , Chae Y.C. , Kim B.G. (1970) **Kang, S., Jang, Y., Chae, Y.C., and Kim, B.G.**
32. Kim B.G. , Gao M.Q. , Choi Y.P. , Kang S. , Park H.R. , Kang K.S. , Cho N.H. (2012) **Invasive breast cancer induces laminin-332 upregulation and integrin $\beta 4$ neoexpression in myofibroblasts to confer an anoikis-resistant phenotype during tissue remodeling** *Breast cancer research : BCR* **14**
<https://doi.org/10.1186/bcr3203>
33. Cherfils J. , Zeghouf M. (2013) **Regulation of small GTPases by GEFs, GAPs, and GDIs** *Physiological reviews* **93**:269–309
<https://doi.org/10.1152/physrev.00003.2012>
34. Schmidt V.A. (2012) **Watch the GAP: Emerging Roles for IQ Motif-Containing GTPase-Activating Proteins IQGAPs in Hepatocellular Carcinoma** *International journal of hepatology* **2012**
<https://doi.org/10.1155/2012/958673>
35. Smith J.M. , Hedman A.C. , Sacks D.B. (2015) **IQGAPs choreograph cellular signaling from the membrane to the nucleus** *Trends in cell biology* **25**:171–184
<https://doi.org/10.1016/j.tcb.2014.12.005>

36. Gorisse L. , Li Z. , Wagner C.D. , Worthylake D.K. , Zappacosta F. , Hedman A.C. , Annan R.S. , Sacks D.B. (2020) **Gorisse, L., Li, Z., Wagner, C.D., Worthylake, D.K., Zappacosta, F., Hedman, A.C., Annan, R.S., and Sacks, D.B. (2020). Ubiquitination of the scaffold protein IQGAP1 diminishes its interaction with and activation of the Rho GTPase CDC42. 295, 4822-4835. 10.1074/jbc.RA119.011491. Ubiquitination of the scaffold protein IQGAP1 diminishes its interaction with and activation of the Rho GTPase CDC42 295:4822-4835**
<https://doi.org/10.1074/jbc.RA119.011491>
37. Woods D. , Turchi J.J. (2013) **Chemotherapy induced DNA damage response: convergence of drugs and pathways** *Cancer biology & therapy* **14**:379-389
<https://doi.org/10.4161/cbt.23761>
38. Cheung-Ong K. , Giaever G. , Nislow C. (2013) **DNA-damaging agents in cancer chemotherapy: serendipity and chemical biology** *Chemistry & biology* **20**:648-659
<https://doi.org/10.1016/j.chembiol.2013.04.007>
39. Kim C. , Gao R. , Sei E. , Brandt R. , Hartman J. , Hatschek T. , Crosetto N. , Foukakis T. , Navin N.E. (2018) **Chemoresistance Evolution in Triple-Negative Breast Cancer Delineated by Single-Cell Sequencing** *Cell* **173**:879-893
<https://doi.org/10.1016/j.cell.2018.03.041>
40. Noordermeer S.M. , van Attikum H. (2019) **PARP Inhibitor Resistance: A Tug-of-War in BRCA-Mutated Cells** *Trends in cell biology* **29**:820-834
<https://doi.org/10.1016/j.tcb.2019.07.008>
41. Geenen J.J.J. , Linn S.C. , Beijnen J.H. , Schellens J.H.M. (2018) **PARP Inhibitors in the Treatment of Triple-Negative Breast Cancer** *Clinical pharmacokinetics* **57**:427-437
<https://doi.org/10.1007/s40262-017-0587-4>
42. Lee A. , Djamgoz M.B.A. (2018) **Triple negative breast cancer: Emerging therapeutic modalities and novel combination therapies** *Cancer treatment reviews* **62**:110-122
<https://doi.org/10.1016/j.ctrv.2017.11.003>
43. Campbell P.S. , Mavingire N. , Khan S. , Rowland L.K. , Wooten J.V. , Opoku-Agyeman A. , Guevara A. , Soto U. , Cavalli F. , Loaiza-Pérez A.I. , et al. (2018) **Campbell, P.S., Mavingire, N., Khan, S., Rowland, L.K., Wooten, J.V., Opoku-Agyeman, A., Guevara, A., Soto, U., Cavalli, F., Loaiza-Pérez, A.I., et al. (2018). AhR ligand aminoflavone suppresses α6-integrin-Src-Akt signaling to attenuate tamoxifen resistance in breast cancer cells. 234, 108-121. 10.1002/jcp.27013. AhR ligand aminoflavone suppresses α6-integrin-Src-Akt signaling to attenuate tamoxifen resistance in breast cancer cells 234:108-121**
<https://doi.org/10.1002/jcp.27013>
44. Ali H.R. , Dawson S.J. , Blows F.M. , Provenzano E. , Pharoah P.D. , Caldas C. (2011) **Cancer stem cell markers in breast cancer: pathological, clinical and prognostic significance** *Breast cancer research : BCR* **13**
<https://doi.org/10.1186/bcr3061>

45. Friedland J.C. , Lakins J.N. , Kazanietz M.G. , Chernoff J. , Boettiger D. , Weaver V.M. (2007) **alpha6beta4 integrin activates Rac-dependent p21-activated kinase 1 to drive NF-kappaB-dependent resistance to apoptosis in 3D mammary acini** *Journal of cell science* **120**:3700–3712
<https://doi.org/10.1242/jcs.03484>
46. Wei T. , Lambert P.F. (2021) **Wei, T., and Lambert, P.F. (2021). Role of IQGAP1 in Carcinogenesis. 13. 10.3390/cancers13163940. Role of IQGAP1 in Carcinogenesis 13**
<https://doi.org/10.3390/cancers13163940>
47. Huang L. , Xu S. , Hu D. , Lu W. , Xie X. , Cheng X. (2015) **IQGAP1 Is Involved in Enhanced Aggressive Behavior of Epithelial Ovarian Cancer Stem Cell-Like Cells During Differentiation** *International journal of gynecological cancer : official journal of the International Gynecological Cancer Society* **25**:559–565
<https://doi.org/10.1097/igc.0000000000000394>
48. Deb S. , Felix D.A. , Koch P. (2021) **Deb, S., Felix, D.A., and Koch, P. (2021). Tnfaip2/exoc3-driven lipid metabolism is essential for stem cell differentiation and organ homeostasis. 22, e49328. 10.15252/embr.201949328. Tnfaip2/exoc3-driven lipid metabolism is essential for stem cell differentiation and organ homeostasis 22**
<https://doi.org/10.15252/embr.201949328>
49. Barutta F. , Bellini S. (2022) **Barutta, F., and Bellini, S. (2022). Protective effect of the tunneling nanotube-TNFAIP2/M-sec system on podocyte autophagy in diabetic nephropathy. 1-20. 10.1080/15548627.2022.2080382. Protective effect of the tunneling nanotube-TNFAIP2/M-sec system on podocyte autophagy in diabetic nephropathy :1–20**
<https://doi.org/10.1080/15548627.2022.2080382>
50. Zhou L. , Niu Z. , Wang Y. , Zheng Y. , Zhu Y. , Wang C. , Gao X. , Gao L. , Zhang W. (2022) **Zhou, L., Niu, Z., Wang, Y., Zheng, Y., Zhu, Y., Wang, C., Gao, X., Gao, L., and Zhang, W. (2022). Senescence as a dictator of patient outcomes and therapeutic efficacies in human gastric cancer. 8, 13. 10.1038/s41420-021-00769-6. Senescence as a dictator of patient outcomes and therapeutic efficacies in human gastric cancer 8**
<https://doi.org/10.1038/s41420-021-00769-6>
51. Guo F. , Yuan Y. (2020) **Guo, F., and Yuan, Y. (2020). Tumor Necrosis Factor Alpha-Induced Proteins in Malignant Tumors: Progress and Prospects. 13, 3303–3318. 10.2147/ott.s241344. Tumor Necrosis Factor Alpha-Induced Proteins in Malignant Tumors: Progress and Prospects 13:3303–3318**
<https://doi.org/10.2147/ott.s241344>
52. Liu Y. , Ji X. , Kang N. , Zhou J. , Liang X. , Li J. , Han T. , Zhao C. , Yang T. (2020) **Tumor necrosis factor α inhibition overcomes immunosuppressive M2b macrophage-induced bevacizumab resistance in triple-negative breast cancer** *Cell death & disease* **11**
<https://doi.org/10.1038/s41419-020-03161-x>
53. Chen C. , Sun X. , Ran Q. , Wilkinson K.D. , Murphy T.J. , Simons J.W. , Dong J.T. (2005) **Ubiquitin-proteasome degradation of KLF5 transcription factor in cancer and untransformed epithelial cells** *Oncogene* **24**:3319–3327
<https://doi.org/10.1038/sj.onc.1208497>

54. Chen C. , Zhou Z. , Ross J.S. , Zhou W. , Dong J.T. (2007) **The amplified WWP1 gene is a potential molecular target in breast cancer** *International journal of cancer* **121**:80–87
<https://doi.org/10.1002/ijc.22653>

Author information

Huan Fang

Key Laboratory of Animal Models and Human Disease Mechanisms, Kunming Institute of Zoology, Chinese Academy of Sciences. Kunming, Yunnan, China., Kunming College of Life Sciences, University of Chinese Academy Sciences, Kunming, Yunnan, China.

Wenlong Ren

Key Laboratory of Animal Models and Human Disease Mechanisms, Kunming Institute of Zoology, Chinese Academy of Sciences. Kunming, Yunnan, China., School of Life Science, University of Science & Technology of China. Hefei, Anhui, China.

Qiuxia Cui

Affiliated Hospital of Guangdong Medical University. Zhanjiang, Guangdong, China., Department of Breast Surgical Oncology, National Cancer Center/National Clinical Research Center for Cancer/Cancer Hospital & Shenzhen Hospital, Chinese Academy of Medical Sciences and Peking Union Medical College, Shenzhen, Guangdong, China., Key Laboratory of Animal Models and Human Disease Mechanisms, Kunming Institute of Zoology, Chinese Academy of Sciences. Kunming, Yunnan, China.
ORCID iD: [0000-0003-3635-5420](https://orcid.org/0000-0003-3635-5420)

Huichun Liang

Key Laboratory of Animal Models and Human Disease Mechanisms, Kunming Institute of Zoology, Chinese Academy of Sciences. Kunming, Yunnan, China.

Chuanyu Yang

Key Laboratory of Animal Models and Human Disease Mechanisms, Kunming Institute of Zoology, Chinese Academy of Sciences. Kunming, Yunnan, China.

Wenjing Liu

Key Laboratory of Animal Models and Human Disease Mechanisms, Kunming Institute of Zoology, Chinese Academy of Sciences. Kunming, Yunnan, China.

Xinye Wang

Key Laboratory of Animal Models and Human Disease Mechanisms, Kunming Institute of Zoology, Chinese Academy of Sciences. Kunming, Yunnan, China.

Xue Liu

Shanghai University of Medicine & Health Sciences, Affiliated Sixth People's Hospital South Campus, Shanghai, China.

Yujie Shi

Department of Pathology, Henan Provincial People's Hospital, Zhengzhou University, Zhengzhou, Henan, 450003, China.

For correspondence: shiyujie523@126.com

Jing Feng

The Second Affiliated Hospital of the Chinese University of Hong Kong (Shenzhen), Shenzhen, 518172, China., Shanghai University of Medicine & Health Sciences, Affiliated Sixth People's Hospital South Campus, Shanghai, China.

For correspondence: fengjing71921@163.com

Ceshi Chen

Academy of Biomedical Engineering, Kunming Medical University, Kunming 650500, China., The Third Affiliated Hospital, Kunming Medical University, Kunming 650118, China., Key Laboratory of Animal Models and Human Disease Mechanisms, Kunming Institute of Zoology, Chinese Academy of Sciences. Kunming, Yunnan, China.

For correspondence: chenc@kmmu.edu.cn

Editors

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Reviewer #1 (Public Review):

In this study, Fang H et al. describe a potential pathway, ITGB4-TNFAIP2-IQGAP1-Rac1, that may involve in the drug resistance in triple negative breast cancer (TNBC). Mechanistically, it was demonstrated that TNFAIP2 bind with IQGAP1 and ITGB4 activating Rac1 and the following drug resistance. The present study focused on breast cancer cell lines with supporting data from mouse model and patient breast cancer tissues. The study is interesting. The experiments were well controlled and carefully carried out. The conclusion is supported by strong evidence provided in the manuscript. The authors may want to discuss the link between ITGB4 and Rac 1, between IQGAP1 and Rac1, and between TNFAIP2 and Rac1 as compared with the current results obtained. This is important considering some recent publications in this area (Cancer Sci 2021, J Biol Chem 2008, Cancer Res 2023).

Reviewer #2 (Public Review):

Breast cancer is the most common malignant tumor in women. One of subtypes in breast cancer is so called triple-negative breast cancer (TNBC), which represents the most difficult subtype to treat and cure in the clinic. Chemotherapy drugs including epirubicin and cisplatin are widely used for TNBC treatment. However, drug resistance remains as a challenge in the clinic. The authors uncovered a molecular pathway involved in chemotherapy drug resistance, and molecular players in this pathway represent as potential drug targets to overcome drug resistance. The experiments are well designed and the conclusions drawn mostly were supported by the data. The findings have potential to be translated into the clinic.

Reviewer #3 (Public Review):

In this manuscript, Fang and colleagues found that IQGAP1 interacts with TNFAIP2, which activates Rac1 to promote drug resistance in TNBC. Furthermore, they found that ITGB4 could interact with TNFAIP2 to promote TNBC drug resistance via the TNFAIP2/IQGAP1/Rac1 axis by promoting DNA damage repair.

This work has good innovation and high potential clinical significance.